

HealthConnect Clinic Experience Lab

Week 7 Project Summary

AnalystLab Africa | Track: Data Analytics

My main goal for Week 7 was to test the accuracy, consistency and usefulness of the analytical work I completed in Week 6.

I started by validating some of my main KPIs and findings against the underlying dataset. The overall no-show rate remained 48.46%. Average booking lead time remained 24.52 days for attended appointments and 34.53 days for no-shows. I also confirmed the reminder-status and previous no-show history results from Week 6.

I tested the Power BI dashboard filters against Excel and confirmed that the filtered dashboard values agreed with the underlying data.

For my main Week 7 analytical test, I combined booking lead time with reminder status. The no-show rate continued to increase as booking lead time became longer in both reminder groups. Appointments where reminders were sent also had lower no-show rates across every booking lead-time group.

Based on this test, I improved the Power BI report by adding a Week 7 Validation & Refinement page showing the combined relationship through a matrix and clustered column chart.

I continued my cross-track collaboration with Data Science by sharing the combined booking lead-time and reminder-status finding for further testing against the model. The finding was independently verified, and an interaction feature combining booking lead time and reminder status was tested in the Logistic Regression model. Accuracy changed from 0.631 to 0.633, while recall for short-lead-time appointments remained at 6.1%, so the interaction feature was not adopted.

One limitation is that the analysis shows associations and does not prove causation. Some deeper comparisons also exclude cancelled appointments, so their percentages use a different denominator from the overall 48.46% no-show rate.

My main contribution this week was validating the existing analytical work, testing the dashboard, refining the analysis and making the relationship between booking lead time and reminder status clearer.

Before Week 8, I will complete the cross-track validation, carry out a final consistency check and prepare the analytical outputs for final integration and presentation.